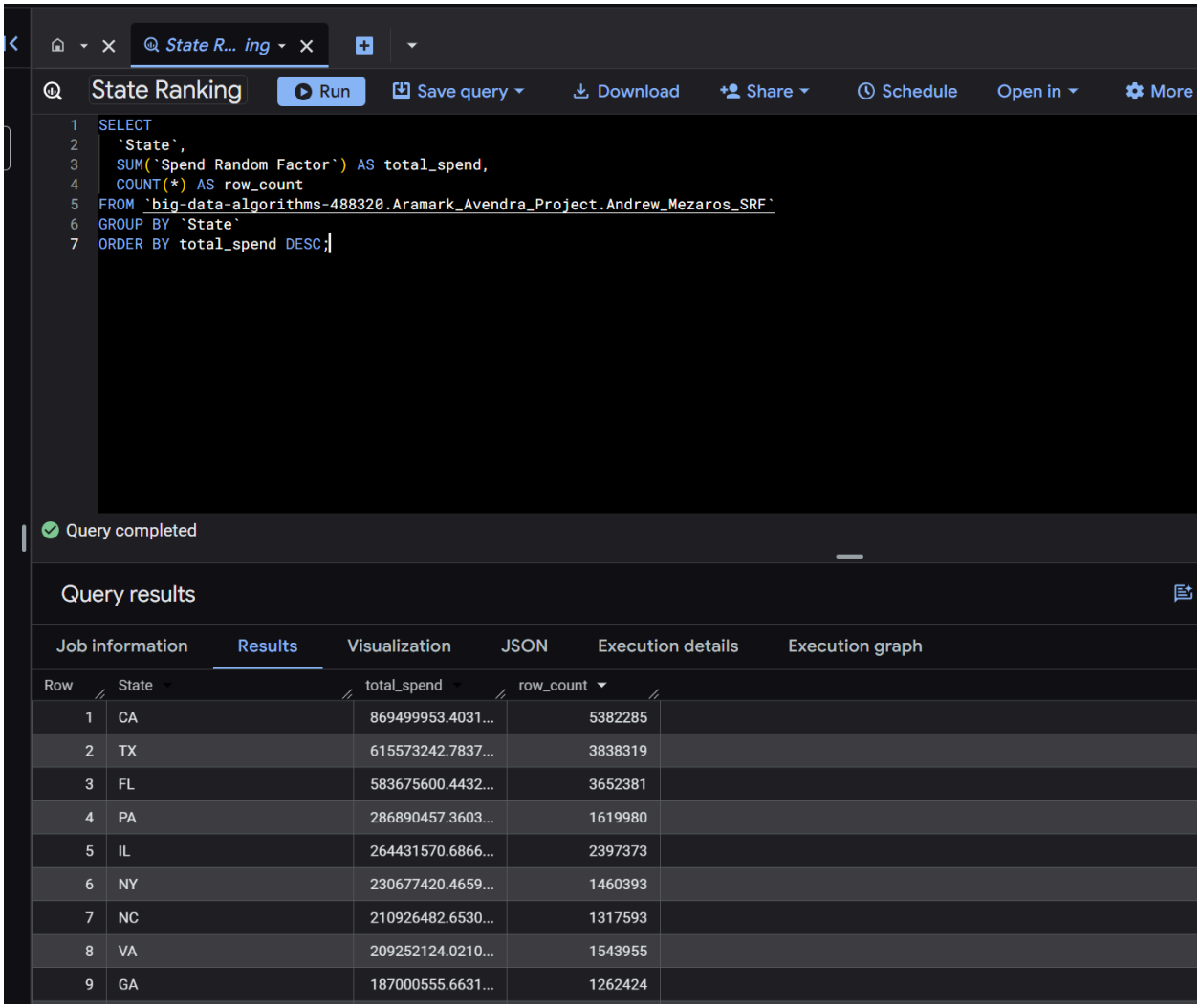
1. 

Row	State	total_spend	row_count
1	CA	869499953.4031...	5382285
2	TX	615573242.7837...	3838319
3	FL	583675600.4432...	3652381
4	PA	286890457.3603...	1619980
5	IL	264431570.6866...	2397373
6	NY	230677420.4659...	1460393
7	NC	210926482.6530...	1317593
8	VA	209252124.0210...	1543955
9	GA	187000555.6631...	1262424

   a. State Ranking:
      i. Establishes baseline geography, and indicates spending over the year per location.
      ii. Helps identify regions where sales might suffer
      iii. Useful when combined with weather to explain reasons for high or low sales

2. State + City Ranking

```google_sql_query
SELECT
  `State`,
  `City`,
  SUM(`Spend Random Factor`) AS total_spend,
  COUNT(*) AS row_count
FROM `big-data-algorithms-488320.Aramark_Avendra_Project.Andrew_Mezaros_SRF`
GROUP BY `State`, `City`
ORDER BY total_spend DESC;
```

Query completed

Query results

Job information | Results | Visualization | JSON | Execution details | Execution graph

Row	State	City	total_spend	row_count
1	TX	HOUSTON	150859659.1729...	483638
2	FL	ORLANDO	109024423.1417...	427864
3	IL	CHICAGO	107095007.3182...	569912
4	PA	PHILADELPHIA	83444709.41891...	297757
5	NY	NEW YORK	82476764.99099...	421592
6	DC	WASHINGTON	60833496.42935...	300999
7	FL	TAMPA	60460798.37874...	255469
8	TN	NASHVILLE	59653101.37387...	246900
9	MA	BOSTON	53813767.13063...	259163

   a. State - City Ranking:
      i. Notifies the high Spending cities within states
      ii. Where there might be population densities for sales
      iii. Reveals whether state spending is localized or distributed
      iv. Useful in the future for anomaly detection

3. Category 1 Breakdown

```google_bigquery_query
SELECT
  `Category Level 1`,
  SUM(`Spend Random Factor`) AS total_spend,
  COUNT(*) AS row_count
FROM `big-data-algorithms-488320.Aramark_Avendra_Project.Andrew_Mezaros_SRF`
GROUP BY `Category Level 1`
ORDER BY total_spend DESC;
```

Query completed

Query results

Job information | Results | Visualization | JSON | Execution details | Execution graph

Row	Category Level 1	total_spend	row_count
1	FOOD	3695921092.883...	29400266
2	BEVERAGE	637775122.6066...	3568234
3	DISPOSABLES	415461473.1056...	3249175
4	MAINTENANCE AND ENGINEER...	381648750.9688...	1599542
5	CHEMICALS AND CLEANING	346193921.75432	2009923
6	ROOM AND SPA	233263025.5060...	1321793
7	FOOD SERVICE EQUIPMENT AN...	157071153.4462...	912109
8	CLOTHING AND FOOTWEAR	81922318.33480...	278806
9	FURNITURE FIXTURES AND EQ...	67229044.22299...	186269

a. Category 1 Breakdown:
   i. Points out what is being purchased at a high level
   ii. Defines priorities for the company
   iii. What should be focused on to improve

4. Category 1 + 2 (Breakdowns)

Run | Save query | Download | Share

```
SELECT
  `Category Level 1` AS category_level_1,
  `Category Level 2` AS category_level_2,
  SUM(`Spend Random Factor`) AS total_spend,
  COUNT(*) AS row_count
FROM `big-data-algorithms-488320.Aramark_Avendra_Project.Andrew_Mezaros_SRF`
GROUP BY `Category Level 1`, `Category Level 2`
ORDER BY category_level_1, total_spend DESC;
```

Query completed

Query results

Job information | Results | Visualization | JSON | Execution details | Execution graph

Row	category_level_1	category_level_2	total_spend	row_count
1	BEVERAGE	COFFEE	137754050.9305...	464659
2	BEVERAGE	NON-CARB, NON-DAIRY	137457001.2638...	658193
3	BEVERAGE	JUICE	131839616.9014...	1117495
4	BEVERAGE	CARBONATED SOFT DRINKS	94379591.28586...	482705
5	BEVERAGE	WATER	82210476.85342...	428182
6	BEVERAGE	BEVERAGES, OTHER	32169436.49656...	99699
7	BEVERAGE	TEA	21945108.92003...	317195
8	BEVERAGE	ICE	19839.95495495...	106

   a. Category 1 + 2 Breakdown
      i. Drives of each category internally
      ii. Useful for ML section to help breakdown the factors of sale
      iii. Points out hidden details to prioritize

5.

Business Entity Type

Run | Save query | Download

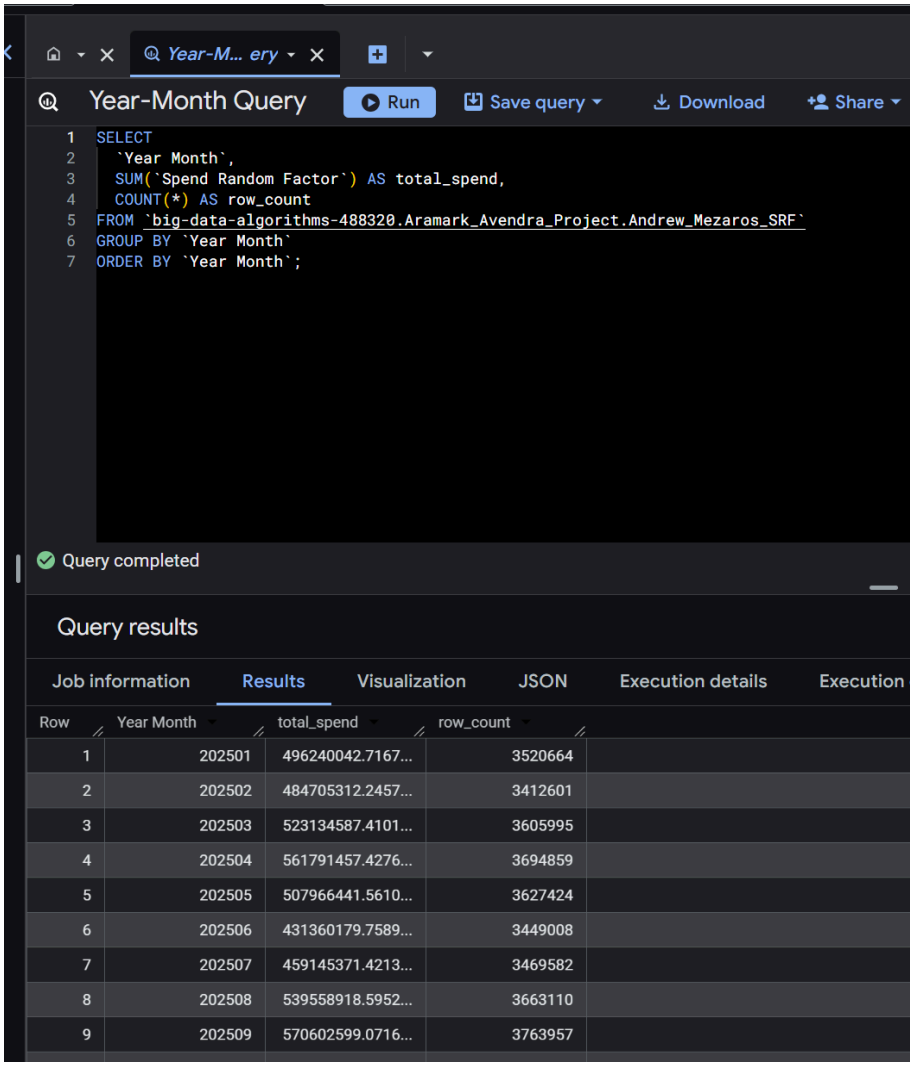
```
SELECT
  `Business Entity Type`,
  SUM(`Spend Random Factor`) AS total_spend,
  COUNT(*) AS row_count
FROM `big-data-algorithms-488320.Aramark_Avendra_Project.Andrew_Mezaros_SRF`
GROUP BY `Business Entity Type`
ORDER BY total_spend DESC;
```

Query completed

Query results

Job information | Results | Visualization | JSON | Execution details

Row	Business Entity Type	total_spend	row_count
1	GPO	3985547316.675...	34298937
2	MANAGED SERVICES	2157628850.954...	8668522

6. 

Row	Year Month	total_spend	row_count
1	202501	496240042.7167...	3520664
2	202502	484705312.2457...	3412601
3	202503	523134587.4101...	3605995
4	202504	561791457.4276...	3694859
5	202505	507966441.5610...	3627424
6	202506	431360179.7589...	3449008
7	202507	459145371.4213...	3469582
8	202508	539558918.5952...	3663110
9	202509	570602599.0716...	3763957

   a. Year- Month Breakdown:
      i. Monthly SRF tracking
      ii. Identify peaks and troughs of seasonal sales

7.

Distributor x State

Run | Save query | Download | Share | Schedule | Open in

```google_sql
SELECT
  `State`,
  `Distributor Group`,
  SUM(`Spend Random Factor`) AS total_spend,
  COUNT(*) AS row_count
FROM `big-data-algorithms-488320.Aramark_Avendra_Project.Andrew_Mezaros_SRF`
GROUP BY `State`, `Distributor Group`
ORDER BY `State`, total_spend DESC;
```

Query completed

Query results

Job information | Results | Visualization | JSON | Execution details | Execution graph

Row	State	Distributor Group	total_spend	row_count
1	AB	FF&E	108.54054054054	1
2	AK	MASTER DISTRIBUTOR	7920226.630630...	53271
3	AK	ENGINEERING MRO	936033.2792792...	5143
4	AK	ROOM OPERATIONS - OTHER	561192.13063063	1837
5	AK	DIRECT BEVERAGE	558050.5720720...	2036
6	AK	*null*	511642.7747747...	167
7	AK	REGIONAL PRODUCE	296933.5810810...	2182
8	AK	ADMINISTRATIVE	286755.2612612...	196

8.

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Category x Business Type

Run | Save query | Download | Share | Schedule | Op

```google_sql
SELECT
  `Business Entity Type`,
  `Category Level 1`,
  SUM(`Spend Random Factor`) AS total_spend,
  COUNT(*) AS row_count
FROM `big-data-algorithms-488320.Aramark_Avendra_Project.Andrew_Mezaros_SRF`
GROUP BY `Business Entity Type`, `Category Level 1`
ORDER BY `Business Entity Type`, total_spend DESC;
```

Query completed

Query results

Job information | Results | Visualization | JSON | Execution details | Execution graph

Row	Business Entity Type	Category Level 1	total_spend	row_count
1	GPO	FOOD	2392718413.148...	22918493
2	GPO	BEVERAGE	289359603.6427...	2666059
3	GPO	CHEMICALS AND CLEANING	280406535.0065...	1782679
4	GPO	DISPOSABLES	276808257.4479...	2718190
5	GPO	ROOM AND SPA	226935669.2582...	1269645

9.

Catego... own

Category 1 x State Breakdown

Run | Save query | Download | Share | Schedul

```
SELECT
  `State`,
  `Category Level 1`,
  SUM(`Spend Random Factor`) AS total_spend,
  COUNT(*) AS row_count
FROM `big-data-algorithms-488320.Aramark_Avendra_Project.Andrew_Mezaros_SRF`
GROUP BY `State`, `Category Level 1`
ORDER BY `State`, total_spend DESC;
```

Query completed

Query results

Job information | Results | Visualization | JSON | Execution details | Execution graph

Row	State	Category Level 1	total_spend	row_count
1	AB	MAINTENANCE AND ENGINEER...	108.54054054054	1
2	AK	FOOD	7104351.283783...	46936
3	AK	MAINTENANCE AND ENGINEER...	1143493.851351...	2840
4	AK	BEVERAGE	1060897.063063...	5769
5	AK	DISPOSABLES	930935.2792792...	5370
6	AK	CHEMICALS AND CLEANING	772448.3468468...	3431
7	AK	ROOM AND SPA	532706.0810810...	1845
8	AK	FOOD SERVICE EQUIPMENT AN...	202072.9954954...	1236
9	AK	RETAIL AND PROMOTIONAL	150640.1531531...	54